

Nardya - the Artin-Schreier Theorem

Lemma (Artin-Schreier 1927)

Let Ω be an alg. closed field, let P be a proper subfield of Ω s.t. Ω/P is finite.

Then P is real closed & $\Omega = P(i)$.

def: a field Ω is real $\Leftrightarrow -1$ is not a sum of squares
 real closed $\Leftrightarrow$ real & no prop alg. ext
 is real.

Kummer extensions & Artin-Schreier extensions: for p prime,

these are cyclic extensions of deg p , either in characteristic p (Artin-Schreier) or in char $\neq p$ for fields containing primitive p^{th} root of unity.

Artin-Schreier polynomial: $X^p - X =: \wp(X)$.

AS-ext & Kummer-ext are building blocks of finite ext.

Pickup: if $P/[K:h]$, $\exists F: \frac{p}{K}$
 $h \quad F \quad K$

Claim: if $\text{char}(K)=p$, any cyclic ext F/K of $\deg p$ is an AS-ext, i.e., $F=K(a)$ when $a^p - a \neq 0$, & vice versa.

Proof: Let β be a root of $X^p - X - a$, $a \in K$. Then $\beta, \beta+1, \dots, \beta+(p-1)$ are all the roots of $X^p - X - a$. So $K(\beta)$ contains all roots of $\phi(X) - a$. if $\beta \notin K$, then $\phi(X) - a$ is irr. over K , so $K(\beta)/K$ is Galois of $\deg p$ & thus cyclic.

Now let $K(\alpha)$ be a cyclic extension of $\deg p$ over K .

Let σ be a generator of $\text{Gal}(K(\alpha)/K)$, i.e., $\text{Gal}(K(\alpha)/K) = \langle \sigma \rangle$.

Let $\alpha_i = \sigma^i(\alpha)$. $A = \begin{pmatrix} 1 & \alpha_0 & \alpha_0^2 & \dots & \alpha_0^{p-1} \\ 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_{p-1} & \alpha_{p-1}^2 & \dots & \alpha_{p-1}^{p-1} \end{pmatrix}$ Vandermonde Matrix

$\det(A) \neq 0 \Leftrightarrow \alpha_i \neq \alpha_j \Rightarrow \exists$ column which doesn't sum to 0: $s = \sum_{i=0}^{p-1} \alpha_i^k \neq 0$ for some k .

Let $\xi = -\frac{1}{s} \sum_{i=0}^{p-1} i \alpha_i^k$. Claim: $K(\xi) = K(a)$ &

ξ is an AS-root.

• $\xi \notin K$, because $\sigma(\xi) = -\frac{1}{s} \sum_{i=0}^{p-1} i \alpha_{i+1}$

(note that $\sigma(s) = s$) $\quad \quad \quad = -\frac{1}{s} \left(\sum_{i=0}^{p-1} (i+1) \alpha_{i+1} - s \right)$

$= \xi + 1 \neq \xi$

So $\xi \notin \text{Fix}(\sigma) = K$

Therefore $K(\xi) \neq K$ & $K(\xi) \subseteq K(\alpha)$,

so $[K(\xi):K] \mid [K(\alpha):K] = p$, so $K(\xi) = K(\alpha)$

• $\xi^p - \xi \in K$ since $\sigma(\xi^p - \xi) = (\xi+1)^p - (\xi+1)$
 $= \xi^p - \xi + 1 - 1$
 $= \xi^p - \xi$

So $\xi \in \text{Fix}(\sigma) = K$

Thm 3.24 in GKS: let $\text{char}(K) = p$. There is

a bijection:

$\{ \text{1-dim subspaces of } \langle c \rangle + p(K) \leq \frac{1}{p(K)} \} \leftrightarrow \{ L/K \text{ Galois ext of type } p \}$

$\langle c \rangle + p(K) \mapsto K(\alpha)$, where $\alpha^p = c$.

- well defined: if $c' \in (\langle c \rangle + \mathfrak{p}(K)) \setminus \mathfrak{p}(K)$, then

$$c' = ac + \mathfrak{p}(b) \text{ with } b \in K, a \in \mathbb{F}_p^\times.$$

Let α be an AS-root of c , set $\beta = a\alpha + b \in K(\alpha)$

we have $\alpha = a^{-1}(\beta - b)$, so $\alpha \in K(\beta)$, so $K(\alpha) = K(\beta)$.

$$\beta^p - \beta = a^p \alpha^p + b^p = a\alpha - b = \overset{c \in \mathbb{F}_p!}{a}(\alpha^p - \alpha) + (b^p - b) = ac + \mathfrak{p}(b) = c'$$

- injective:
- if $K(\alpha) = K(\beta)$, then $\beta = a\alpha + b$ with $a \in \mathbb{F}_p^\times, b \in K$
(why?) $\rightarrow$ AS27, Satz 2

$$\alpha^p - \alpha = c \in K, \beta^p - \beta = d \in K:$$

$$d = \beta^p - \beta = a(\alpha^p - \alpha) + (b^p - b) = ac + \mathfrak{p}(b) \in \langle c \rangle + \mathfrak{p}(K)$$

$$\text{so } \langle c \rangle + \mathfrak{p}(K) = \langle d \rangle + \mathfrak{p}(K).$$

• surjective clear.

Lemma: 3.25 in GKS, Satz 3 in AS27).

Let K be a field of $\text{char}(K) = p > 0$. If $\exists L/K$ of $\text{deg } p$ Galois
then $\exists M/K$ of $\text{deg } p$, M/K of $\text{deg } p^2$ cyclic.

Proof: by hand in AS27, by Hilbert 90 in GKS:

GKS 3.22: L/K finite cyclic, σ a generator of $\text{Gal}(L/K)$, $a \in L$
Hilbert 90 $T_{L/K}(a) = 0 \Leftrightarrow \exists b \in L, a = \sigma(b) - b$ (where $T_{L/K}(a) = \sum_{i=0}^{p-1} \sigma^i(a)$)

$$L = K(\xi), \quad \xi^p - \xi = a \in K \Rightarrow -a^{-1} - a^{-1} \xi^{p-1} = -\xi^{-p} \Leftrightarrow \text{Tr}(\xi^{-1}) = 0$$

$$\text{Gal}(L/K) \cong \langle \sigma \rangle \Rightarrow -\xi^{p-1} = -1 - a \xi^{-1}$$

$$\text{Tr}(\underbrace{-\xi^{p-1}}_{=: b}) = \text{Tr}(-1 - a \xi^{-1}) = \underbrace{-\text{Tr}(1)}_{=0} - a \underbrace{\text{Tr}(\xi^{-1})}_{=-a^{-1}} = 1$$

$$\text{Tr}(b^p - b) = \text{Tr}(b^p) - \text{Tr}(b) = 1 - 1 = 0 \stackrel{\text{H00}}{\Rightarrow} b^p - b = \sigma(c) \cdot c$$

for $c \in L$,
 $c \notin K$ so $\sigma(c) \neq c$

Claim: let $y^p - y = c$, then $M = K(y)$ is an extension.

• c has no AS-rt in L .

if $c = \sigma(e)$, $e \in L$, then

$$\begin{aligned} \sigma(b) &= \sigma(\sigma(e) \cdot c) \\ &= \sigma(\sigma(e)) - \sigma(e) \end{aligned}$$

$$\sigma(b) = \sigma(e) + e = 0$$

$$b - \sigma(e) + e = h \in \mathbb{F}_p$$

$$\Rightarrow b - h = \sigma(e) - e$$

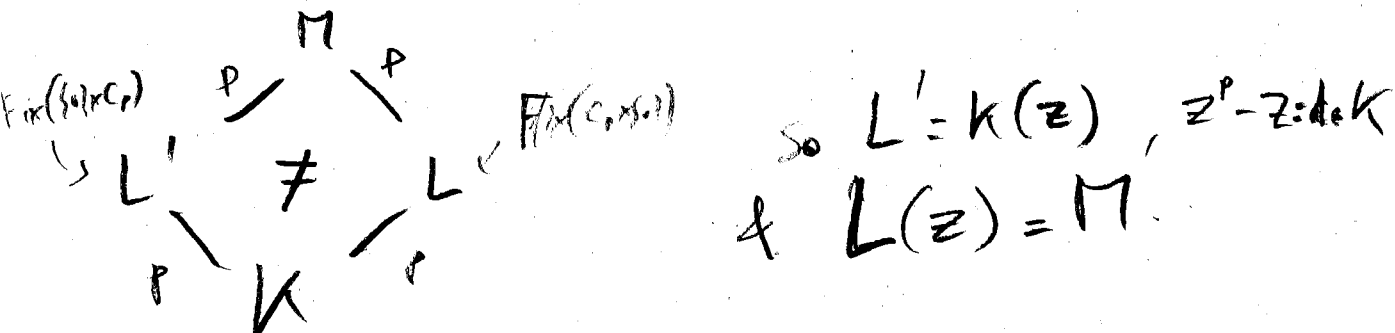
$$\text{H00} \Rightarrow \text{Tr}(b - h) = 0$$

$$\Rightarrow \text{Tr}(b) = 0 \quad \checkmark$$

$$[M:K] = [M:L][L:K] = p^2$$

$$\bullet \text{ So, } \text{Gal}(M/K) \cong \begin{cases} C_{p^2} \\ C_p \times C_p \end{cases}$$

Assume $\text{Gal}(M/K) = C_p \times C_p$, then Galois cor. sp. has



Now

$$\langle c \rangle + p(L) = \langle d \rangle + p(L)$$

$$\text{Since } L(y) = M = L(z)$$

$$\& y^p - y = c, \quad z^p - z = d$$

$$\text{So } c = \text{id} + e^p - e \text{ for } e \in \mathbb{F}_p^x \& e \in L$$

$$(bc \ c \in \langle d \rangle + p(L)) \& b^p - b = \sigma(c) - c$$

$$= \sigma(\text{id} + e^p - e) - \text{id} - e^p + e$$

$$= (\sigma(e) - e)^p - (\sigma(e) - e)$$

$$(bc \ d \in K)$$

$$\text{So } p(b) = p(\sigma(e) - e)$$

$$\Rightarrow p(b - \sigma(e) + e) = 0$$

$$\Rightarrow b - \sigma(e) + e = j \in \mathbb{F}_p$$

$$\Rightarrow \text{Tr}(\sigma(e) - e) = 0 \text{ (H90)}$$

$$\& \text{Tr}(b + j) = \text{Tr}(b) \neq 0 \downarrow$$

Back to the thm: if $\mathbb{F}_p$ is finite (I don't think), then

• P is perfect (easy)

• if $p \nmid |E|$, then $\text{char}(P) \neq p$ (otherwise, $\text{Tr} p^n = 0 \forall n$).

• recovery info about char from $\text{Gal}(P/P)$!!